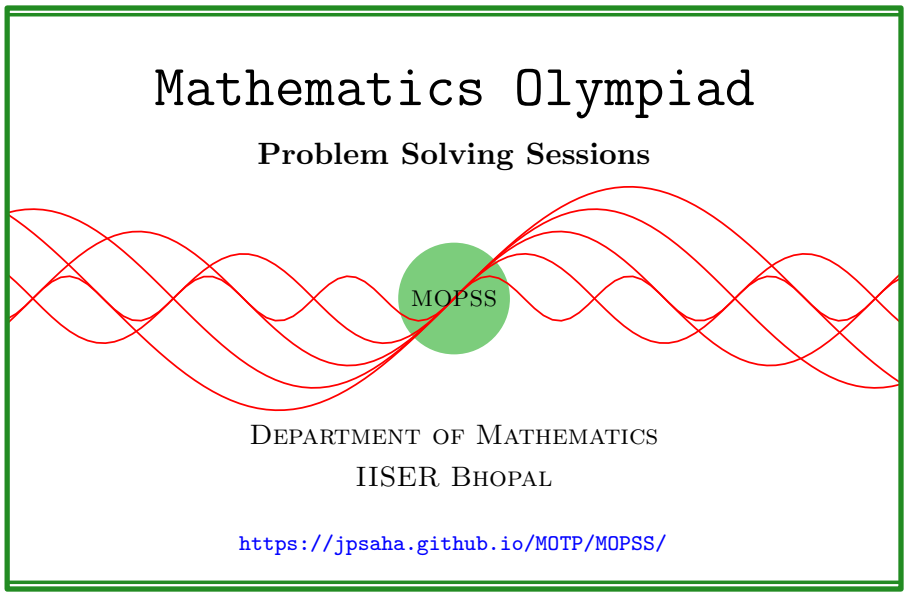
Games

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

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§1 Strategies

Exercise 1 (Covering a mutilated chessboard)**.** Assume that from an 8×8 chessboard, two squares of different colors are removed. Prove that the remaining 62 squares can be covered by 31 dominoes, where each domino covers exactly two adjacent squares.

> **Walkthrough** — Show that there is a trail passing through all the squares of the chessboard, passing through each square exactly once, such that the adjacent squares in the trail are of different colors.

Solution 1. Consider the closed trail as shown in Fig. 1. The trail passes through all 64 squares of the chessboard, and adjacent squares in the trail are of different colors. Removing two squares of different colors from the chessboard breaks the trail into two paths, each of which admits a covering by dominoes. Hence, the remaining 62 squares can be covered by 31 dominoes.

■

Exercise 2 (China TST 1995 Day 2 P2)**.** Alice and Bob play a game with a polynomial of degree at least 4:

$$x^{2n} + \square x^{2n-1} + \square x^{2n-2} + \cdots + \square x + 1.$$

They take turns to fill the empty boxes. If the resulting polynomial has no real root, Alice wins, otherwise, Bob wins. If Alice goes first, who has a winning strategy?

> **Walkthrough** —
>
> **(a)** There are more odds than evens among the integers $1, 2, \ldots, 2n-1$.
>
> **(b)** Can Bob have a winning strategy using the odds in his favour?

Solution 2. Bob has a winning strategy, as described below.

Bob makes sure that at the end of each of his turns except the last one, the number of even powers of x whose coefficients have been provided by some of

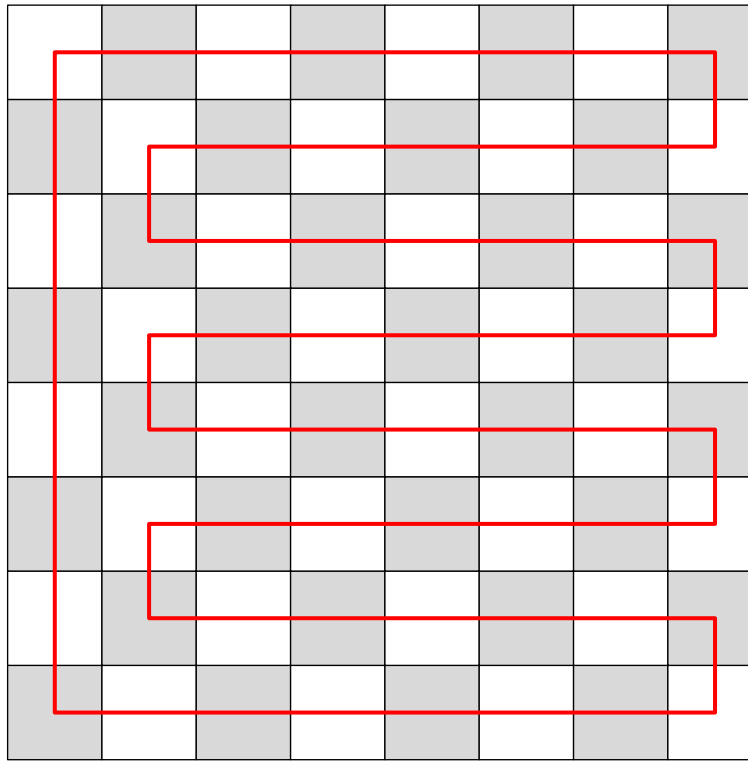


Figure 1: A closed trail through all squares of the chessboard.

The content posted here and at this blog by Evan Chen are quite useful.

them is equal to the number of odd powers of x whose coefficients have been provided by some of them. This can be done, for instance, if during a turn of Bob, other than the last turn, Bob provides the coefficient of an odd (resp. even) power of x if Alice has provided the coefficient of an even (resp. odd) power of x in the preceeding turn.

Since $n \geq 2$, it follows that Bob gets at least one turn. At the beginning of the final turn of Bob, there are two powers of x whose coefficients are to be determined, denote them by x^i, x^j, their coefficients by c_i, c_j respectively. Let $Q(x)$ denote the polynomial, obtained by the erasing the terms corresponding to x^i, x^j from the polynomial that Bob had at the beginning of his final turn. Note that

$$P(x) = Q(x) + c_i x^i + c_j x^j.$$

Note that at least of i, j is odd. Interchanging i, j if necessary, let us assume that i is odd. We descrie the strategy that Bob follows in the two cases below.

Let us consider the case that j is even. Bob determines c_j in such a way that for any choice of c_i, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(1) + Q(-1) + 2c_j = 0.$$

For any choice of c_i, the above choice of c_j shows that $P(1) + P(-1) = 0$, which implies that $P(x)$ has a root in the interval $[-1, 1]$.

Let us consider the case that j is odd. Bob determines c_j in such a way that for any choice of c_i by Alice in the next turn, the completed polynomial $P(x)$ is guaranteed to have at least one real root. This can be done, for instance, by taking c_j satisfying

$$Q(2) + c_j 2^j + 2^i Q(-1) - c_j 2^i = 0.$$

Since $i \neq j$, the above holds for some $c_j \in \mathbb{R}$. For any choice of c_i, note that

$$P(2) + 2^i P(-1) = 0$$

holds, which implies that $P(x)$ has a root in $[-1, 2]$. ■

Exercise 3 (British Mathematical Olympiad Round 1 2003/4 P3)**.** Alice and Barbara play a game with a pack of $2n$ cards, on each of which is written a positive integer. The pack is shuffled and the cards laid out in a row, with the numbers facing upwards. Alice starts, and the girls take turns to remove one card from either end of the row, until Barbara picks up the final card. Each girl's score is the sum of the numbers on her chosen cards at the end of the game. Prove that Alice can always obtain a score at least as great as Barbara's.

Walkthrough —

(a)

Solution 3. to write ■

Exercise 4 (Moldova National Olympiad 2006 Grade 11 P4, , cf. QEDMO 2006 P11)**.** On a table, there are 2006 cards, each with a natural number written on it. Two players play a game, taking turns. In each turn, a player takes a card from either the leftmost or the rightmost end of the row of cards. The game ends when all cards are taken. After the game, each player calculates the sum of the numbers on the cards they have taken. If the sum of the first player is not less than the sum of the second player, then the first player wins. Show that the first player has a winning strategy. (Note that the integers on the cards are visible to both players throughout the game.)

Walkthrough —

(a) Show that the first player can always take the cards in the odd positions or the cards in the even positions.

(b) Before the game starts, the first player compares the sum of the numbers on the cards in the odd positions and the sum of the numbers on the cards in the even positions.

Solution 4. The first player can use the following strategy to win the game. Before the game starts, the first player calculates the sum of the numbers on the cards in the odd positions, and the sum of the numbers on the cards in the even positions. Since there are 2006 cards, one of these sums is greater than or equal to the other. If the sum of the numbers on the cards in the odd positions is greater than or equal to the sum of the numbers on the cards in the even positions, then the first player always takes the cards in the odd positions. Otherwise, the first player always takes the cards in the even positions. In either case, the first player wins.

Note that the first player can always take the cards in the odd positions or the cards in the even positions, because the first player starts the game and there are an even number of cards. Indeed, if the first player takes the card in the odd (resp. even) position in the first turn from the left (resp. right) end of the row, then after each turn of the second player, the first player can respond by taking the card adjacent to the card taken by the second player, which is in the odd (resp. even) position. Thus, the first player can always take the cards in the odd positions or the cards in the even positions, and hence ensures that the sum of the numbers on the cards taken by the first player is greater than or equal to the sum of the numbers on the cards taken by the second player. Consequently, the first player wins. ■

Exercise 5 (Bulgaria National Olympiad 2010 Day 1 P1,). A table 2×2010 is divided to unit cells. Ivan and Peter are playing the following game. Ivan starts, and puts horizontal 1×2 domino that covers exactly two unit table cells. Then Peter puts vertical 2×1 domino that covers exactly two unit table cells. Then Ivan puts horizontal 1×2 domino. Next Peter puts vertical 2×1 domino, and so on. The player who cannot put a domino loses. Which player has a winning strategy?

> **Walkthrough** —
>
> **(a)**

Solution 5. We claim that Ivan, the first player, has a winning strategy.

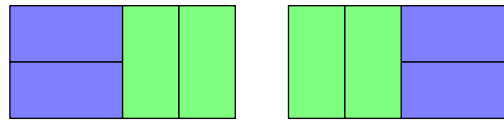
Note that the 2×2010 table can be divided into its leftmost 2×2 block, together with 502 blocks consisting of 2×4 rectangles. Denote the leftmost 2×2 block by A, and the 502 blocks by $B_1, B_2, \ldots, B_{502}$ from left to right. Then Ivan's winning strategy is as follows:

- In the first move, Ivan puts a horizontal domino in A.
- After that, if Peter puts a vertical domino in B_i for the first time for some i (for instance, a green domino as in Fig. 2a or in Fig. 2b), then Ivan puts a horizontal domino in B_i (for instance, a blue domino as in Fig. 2a or in Fig. 2b) such that another horizontal domino and a vertical domino can be put in B_i after that.
- If Peter puts a vertical domino in B_i for the second time for some i, then Ivan puts a horizontal domino in B_i.

Observe that the third step of the strategy is well-defined, since if Peter puts a vertical domino in B_i for the second time, then Ivan does have the option to put a horizontal domino in B_i by the second step of the strategy. Note that after Ivan's first move, Peter cannot put a vertical domino in A. Also, if Peter puts a vertical domino in B_i for the first time, then Ivan can put a horizontal domino in B_i following the second step of the strategy. If Peter puts a vertical domino in B_i for the second time, then Ivan can put a horizontal domino in B_i following the third step of the strategy. Thus, after every move of Peter, Ivan can follow the strategy to put a horizontal domino in some block B_i. Therefore, Peter will run out of moves before Ivan does, and hence Ivan has a winning strategy.

■

Exercise 6 (Junior Balkan MO Shortlist 2015 C1,). An $n \times n$ board (with $n \geq 3$) is divided into n^2 unit squares. Integers from 0 to n included, are written down: one integer in each unit square, in such a way that the sums of integers in each 2×2 square of the board are different. Find all n for which such boards exist.



(a) (b)

Figure 2: Bulgaria National Olympiad 2010 Day 1 P1, Exercise 5

Walkthrough —

(a)

Solution 6. Note that there are $(n-1)^2$ different 2×2 squares in the board, and the sum of the integers in each 2×2 square is at least 0 and at most $4n$. Thus, if n is such that $(n-1)^2 > 4n+1$, then by the pigeonhole principle, it would follow that there are two 2×2 squares with the same sum, and thus, no such board can exist. Hence, we must have $(n-1)^2 \leq 4n+1$, which is equivalent to $n^2 \leq 6n$. Using the fact that $n \geq 3$, we conclude that $3 \leq n \leq 6$.

Fig. 3 shows that such boards exist for $n = 3$, Fig. 4 shows that such boards exist for $n = 4$, and Fig. 5 shows that such boards exist for $n = 5$ and $n = 6$.

This proves that such boards exist precisely for $n = 3, 4, 5, 6$. ■

0	0	0
0	0	1
1	2	3

(a)

0	1	2
0	0	0
0	2	2

(b)

0	1	2
0	0	0
0	0	2

(c)

Figure 3: Junior Balkan MO Shortlist 2015 C1, Exercise 6, for $n = 3$

Exercise 7 (European Mathematical Cup 2016 Senior Category P2, proposed by Petar Orlić)**.** For two positive integers a and b, Ivica and Marica play the following game: Given two piles of a and b cookies, on each turn a player takes $2n$ cookies from one of the piles (with $n \geq 1$), of which he eats n and puts n of them on the other pile. Number n is arbitrary in every move. Players take turns alternatively, with Ivica going first. The player who cannot make a move, loses. Assuming both players play perfectly, determine all pairs of numbers (a, b) for which Marica has a winning strategy.

It is worth noting the apparent similarity of the above problem to Exercise 11.

The content posted here and at this blog by Evan Chen are quite useful.

0	0	0	0
0	0	1	1
1	2	2	3
1	0	2	1

(a)

4	4	4	4
0	0	2	2
0	0	0	0
0	1	2	3

(b)

Figure 4: Junior Balkan MO Shortlist 2015 C1, Exercise 6, for $n = 4$

5	5	5	5	5
0	0	2	2	4
0	0	0	0	0
0	1	2	3	4
5	5	5	5	5

(a)

6	6	6	6	5	5
6	6	5	5	5	5
0	0	2	2	4	4
0	0	0	0	0	0
0	1	2	3	4	5
6	3	5	5	5	5

(b)

Figure 5: Junior Balkan MO Shortlist 2015 C1, Exercise 6, for $n = 5$ and $n = 6$

Walkthrough —

(a)

Solution 7. We claim that Marica has a winning strategy if and only if $|a - b| \leq 1$.

Claim — If $|a - b| > 1$, then Ivica has a move so that the number of cookies in the resulting piles differ by at most 1. If $|a - b| \leq 1$, then any move of Ivica results in two piles of cookies that differ by more than 1 cookie.

Proof of the claim. Let us first consider the case that $|a - b| > 1$. Then there exists a nonzero integer n such that $|a-b-3n|$ lies in $\{-1, 0, 1\}$. If n is positive, then noting that

$$a - 2n \geq b + n - 1 \geq 0,$$

Ivica takes $2n$ cookies from the pile with a cookies, eats n of them and puts n on the pile with b cookies. If n is negative, then noting that

$$b + 2n \geq a - n - 1 \geq 0,$$

Ivica takes $-2n$ cookies from the pile with b cookies, eats $-n$ of them and puts $-n$ on the pile with a cookies. In either case, the number of cookies in the resulting piles differ by at most 1.

Now let us consider the case that $|a-b| \leq 1$. Then for any nonzero integer n, we have

$$|a-b-3n| \geq |3n| - |a-b| \geq 3-1 = 2.$$

Thus any move of Ivica results in two piles of cookies that differ by more than 1 cookie. □

Claim — Marica has a winning strategy if and only if $|a-b| \leq 1$.

Proof of the claim. If $a+b$ is equal to 1, then Marica wins. If $a=1, b=1$, then Marica wins. If a pile has two cookies and the other pile has none, then Ivica has a move which results in two piles, one with 1 cookie and the other with none, and Marica loses. This proves that if $a+b \leq 2$, then Marica has a winning strategy if and only if $|a-b| \leq 1$.

Let $n \geq 2$ be an integer such that if $a+b \leq n$, then Marica has a winning strategy if and only if $|a-b| \leq 1$.

Consider the case that $a+b = n+1$. If $|a-b| \leq 1$, then by the previous claim, any move of Ivica results in two piles of cookies that differ by more than 1 cookie, and by the induction hypothesis, Marica has a winning strategy. If $|a-b| > 1$, then by the previous claim, Ivica has a move so that the number of cookies in the resulting piles differ by at most 1, and by the induction hypothesis, Ivica has a winning strategy, so Marica does not have a winning strategy.

By induction, Marica has a winning strategy if and only if $|a-b| \leq 1$. □

■

Exercise 8 (Canadian Mathematical Olympiad 2016 P1,). The integers $1, 2, 3, \ldots, 2016$ are written on a board. You can choose any two numbers on the board and replace them with their average. For example, you can replace 1 and 2 with 1.5, or you can replace 1 and 3 with a second copy of 2. After 2015 replacements of this kind, the board will have only one number left on it.

(i) Prove that there is a sequence of replacements that will make the final number equal to 2.

(ii) Prove that there is a sequence of replacements that will make the final number equal to 1000.

Walkthrough —

(a)

The content posted here and at this blog by Evan Chen are quite useful.

Solution 8.

> **Claim —** Let x, y be positive integers with $y - x \geq 2$. There is a sequence of moves, when performed to the integers $x, x+1, \ldots, y$, results in the final number $x + 1$. Moreover, there is a sequence of moves, when performed to the integers $x, x + 1, \ldots, y$, results in the final number $y - 1$.

Proof of the claim. If $y - x = 2$, then the claim is true since we can replace x and y with their average, and next, replace the average and $x + 1$ with their average to get $x + 1$, which is equal to $y - 1$.

Assume that $t \geq 2$ be an integer such that the claim is true for all pairs of integers x, y with $y - x \leq t$. Let x, y be positive integers with $y - x = t + 1$. By the induction hypothesis, there is a sequence of moves, when performed to the integers $x+1, x+2, \ldots, y$, results in the final number $x+2$. Next, replace x and $x + 2$ with their average to get $x + 1$. Moreover, by the induction hypothesis, there is a sequence of moves, when performed to the integers $x, x+1, \ldots, y-1$, results in the final number $y - 2$. Next, replace $y - 2$ and y with their average to get $y - 1$.

By induction, the claim is true for all pairs of integers x, y with $y - x \geq 2$. □

Apply the claim with $x = 1$ and $y = 2016$ to get the final number 2.

Apply the claim with $x = 1$ and $y = 999$ to get the final number 998. Next, apply the claim with $x = 1000$ and $y = 2016$ to get the final number 1001. Finally, replace 998 and 1001 with their average to get 1000. ■

Exercise 9 (RMO 2016e P6,). A deck of 52 cards is given. There are four suites each having cards numbered $1, 2, \ldots, 13$. The audience chooses some five cards with distinct numbers written on them. The assistant of the magician comes by, looks at the five cards and turns exactly one of them face down and arranges all five cards in some order. Then the magician enters and with an agreement made beforehand with the assistant, he has to determine the face down card (both suite and number). Explain how the trick can be completed.

Solution 9. The following could be an agreement between the magician and the assistant, so that the magician can determine the face down card.

1. Given a collection of five cards, having distinct numbers written on them, the assistant puts them in a row, as follows.

2. Note that among the five cards, at least two are of the same suite.

3. The assistant chooses two cards of the same color, and puts them next to each other, and then keeps the remaining three cards to their right.

4. Note that if the integers $1, 2, \ldots, 13$ are written along the circumference of a circle Γ (in some order that is known to both the magician and

the assistant, for instance, in the clockwise direction), then for any two distinct integers i, j lying between 1 and 13, one of them (call it Q) can be reached from the other one (call it P) by traversing from P along the circumference in the clockwise or in the anti-clockwise direction within at most 6 steps.

5. Denote the chosen cards of the same color, by P and Q, so that Q can be reached from P, by traversing along the circumference, in the clockwise or in the anti-clockwise direction, within at most 6 steps. The assistant puts the cards P and Q in the order Q, P (resp. P, Q) if Q can be reached from P, by traversing along the circumference, in the anti-clockwise (resp. clockwise) direction, in $s \leq 6$ steps. The assistant turns the card Q face down.

6. The assistant keeps the remaining three cards to the right of these two cards, in an order, which helps to determine the integer s as follows.

7. Denote the integers written on the remaining three cards by a, b, c with $a < b < c$. The $3! = 6$ permutations of $1, 2, 3$ can be identified with the integers $1, 2, 3, 4, 5, 6$ in some order (known to both the magician and the assistant), and the $3! = 6$ permutations of the three remaining cards can be identified with the permutations of $1, 2, 3$ in an obvious way that is known to both the magician and the assistant (for instance, sending a permutation σ of $\{a, b, c\}$ to the permutation $f \circ \sigma \circ f^{-1}$ of $\{1, 2, 3\}$ where $f : \{a, b, c\} \to \{1, 2, 3\}$ denotes the map sending a, b, c to $1, 2, 3$ respectively.). Under the composition of these two identifications, a positive integer ≤ 6 corresponds to a unique permutation of the three remaining cards. The assistant keeps the three remaining cards to the right of the first two cards, as per the permutation which corresponds to the integer s.

Since the suite of the face down card is same as the face up card which is the first in the row of the five cards, the magician determines the suite of the face down card. The magician knows that the number written on the face down card can be reached, by traversing along Γ from the number written on the first face up card, in the clockwise (resp. anti-clockwise) direction if the face down card in the second (resp. first) one in the row of five. ■

Exercise 10 (SMMC 2017 A1,). The five sides and five diagonals of a regular pentagon are drawn on a piece of paper. Two people play a game, in which they take turns to colour one of these ten line segments. The first player colours line segments blue, while the second player colours line segments red. A player cannot colour a line segment that has already been coloured. A player wins if they are the first to create a triangle in their own colour, whose three vertices are also vertices of the regular pentagon. The game is declared a draw if all ten line segments have been coloured without a player winning.

Determine whether the first player, the second player, or neither player can force a win.

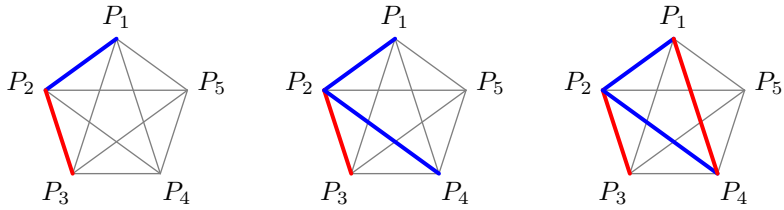


Figure 6: SMMC 2017 A1, Exercise 10, First two colored segments share a common vertex

Walkthrough — To force a win, the first player colors a side of the pentagon in blue in the first move.

Let us first consider the case that the two edges coloured in the first two moves have a common endpoint, as in Fig. 6.

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_2P_3 in red, where $P_1P_2P_3P_4P_5$ denotes the pentagon.

(b) Next, the first player colours P_2P_4 in blue.

(c) If the second player does not color P_1P_4, then in the next move the first player colours P_1P_4 in blue and wins. It remains to consider the case that the second player colours P_1P_4 in red, which we assume from now on.

(d) The first player colours P_2P_5 in blue.

(e) During the next turn of the first player, one of the edges P_1P_5 and P_4P_5 is coloured in blue, thus creating a blue triangle.

We are still left with the case that the two edges coloured in the first two moves do not have a common endpoint, as in Fig. 7.

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_3P_4 in red.

(b) Next, the first player colours P_1P_5 in blue.

(c) If the second player does not color P_2P_5, then in the next move the first player colours P_2P_5 in blue and wins. It remains to consider the case that the second player colours P_2P_5 in red, which we assume from now on.

(d) The first player colours P_1P_3 in blue.

(e) During the next turn of the first player, one of the edges P_2P_3 and P_3P_5 is coloured in blue, thus creating a blue triangle.

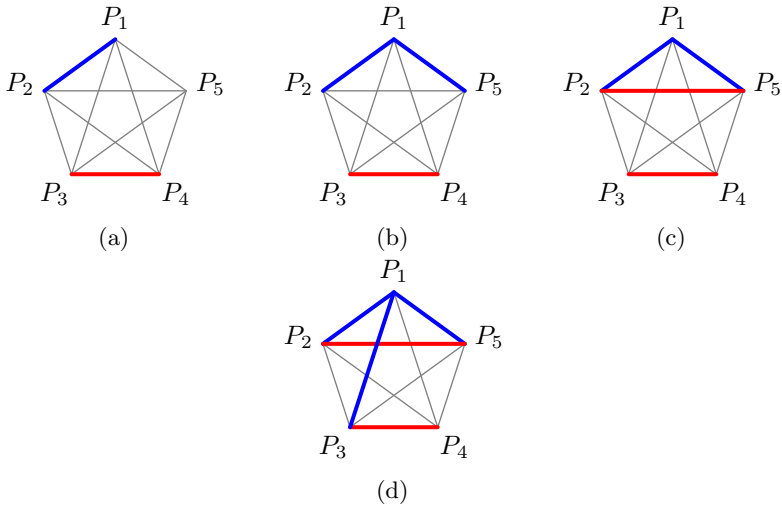


Figure 7: SMMC 2017 A1, Exercise 10, First two colored segments share no common vertex

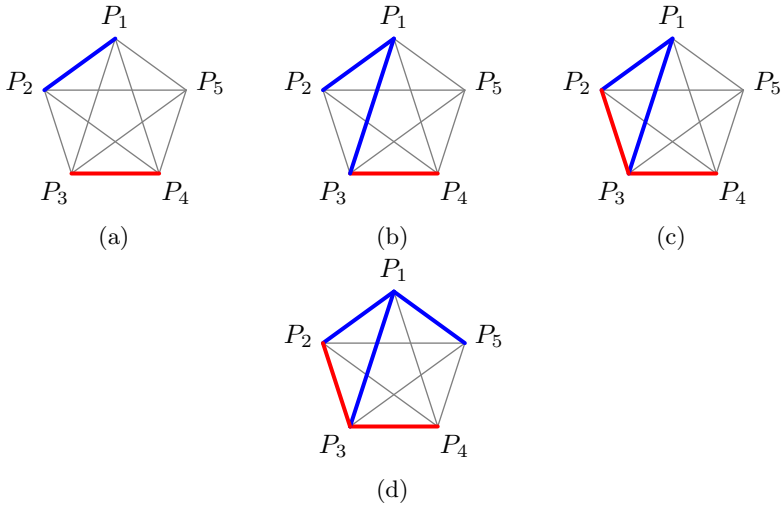


Figure 8: SMMC 2017 A1, Exercise 10, First two colored segments share no common vertex

Here is another strategy to see that the first player can force a win, if the two edges coloured in the first two moves do not have a common endpoint, as in Fig. 8.

(a) Show that there is no loss of generality in assuming that the first player colours P_1P_2 in blue and the second player colours P_3P_4 in red.

(b) Next, the first player colours P_1P_3 in blue.

(c) This forces the second player to colour P_2P_3 in red, otherwise the first player colours P_2P_3 in blue and wins.

(d) The first player colours P_1P_5 in blue.

(e) During the next turn of the first player, one of the edges P_2P_5 and P_3P_5 is coloured in blue, thus creating a blue triangle.

Solution 10. to write ■

Exercise 11 (Balkan Mathematical Olympiad 2018 P3, , proposed by Dimitris Christophides, Cyprus)**.** Alice and Bob play the following game: They start with non-empty piles of coins. Taking turns, with Alice playing first, each player choose a pile with an even number of coins and moves half of the coins of this pile to the other pile. The game ends if a player cannot move, in which case the other player wins.

Determine all pairs (a, b) of positive integers such that if initially the two piles have a and b coins respectively, then Bob has a winning strategy.

It is worth noting the apparent similarity of the above problem to Exercise 7.

Walkthrough —

(a)

Solution 11. We prove that Bob has a winning strategy if and only if a, b are both even and have the same 2-adic valuations, and this common 2-adic valuation is even.

For a positive integer n, let $v_2(n)$ denote the largest integer k such that 2^k divides n, and it is called the 2-adic valuation of n.

Claim — If, prior to the turn of a player, the number of coins in the piles are odd, then the other player wins.

Proof of the Claim. If both piles have an odd number of coins, then the current player cannot make a move, and hence, the other player wins. □

Claim — If at some stage of the game, the number of coins in one of the piles is even and the other pile has an odd number of coins, then the game continues indefinitely, and hence, none of the players has a winning

strategy.

Proof of the Claim. If at some stage of the game, one pile has an even number of coins and the other has an odd number of coins, then the current player is forced to choose the pile with an even number of coins. After moving half of the coins from the even pile to the odd pile, the resulting piles will again consist of one pile with an even number of coins and the other pile with an odd number of coins. By repeating the same argument, more precisely, by induction on the number of turns, it follows that the game continues indefinitely, and hence, none of the players has a winning strategy. □

Claim — If a, b have the same 2-adic valuations, then Bob is forced to win if and only if $v_2(a)$ is even.

Proof of the Claim. If a, b have the same 2-adic valuations, then after each turn, the resulting piles will also have the same 2-adic valuations, which is one less than the 2-adic valuations of a, b. Therefore, the game continues until both piles have an odd number of coins, and then the next player to move loses by the first claim. Note that if $v_2(a)$ is even (resp. odd), then both the piles will have an odd number of coins after an even (resp. odd) number of turns, and hence, both the piles will have an odd number of coins after some turn of Bob (resp. Alice), and consequently, Bob (resp. Alice) is forced to win. This completes the proof of the claim. □

Claim — Assume that a, b are even, and are of different 2-adic valuations. If the smaller 2-adic valuation among a, b is odd, then Alice has a winning strategy.

Proof of the Claim. Alice follows the strategy to choose the pile with the smaller 2-adic valuation in her first turn, and after her turn, the resulting piles will have the same 2-adic valuation, which is one less than the smaller 2-adic valuation among a, b, and is even. Now, Bob and Alice will alternate in taking turns, and by the previous claim, it follows that the second player, which is now Alice, is forced to win. □

Claim — If at some stage of the game, the numbers of coins in the piles are even, and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is even, then after the next turn, the numbers of coins in the resulting piles are even, and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the resulting piles is equal to the smaller of the 2-adic valuations of the numbers of coins in the piles, prior to the turn of

the current player, and consequently, the game continues indefinitely, and hence, none of the players has a winning strategy.

Proof of the Claim. Let x, y be the numbers of coins in the piles at some stage of the game. Assume that x, y are even and $v_2(x) < v_2(y)$. If the current player chooses the pile with x coins, then after his/her turn, the resulting piles will have the same 2-adic valuation, which is $v_2(x) - 1$, and is odd, and hence, by the previous claim, the current player is forced to lose. Therefore, the current player is forced to choose the pile with y coins. If $v_2(y) \geq v_2(x) + 2$, then after the next turn, the numbers of coins in the resulting piles are even and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is equal to $v_2(x)$, which is even. If $v_2(y) = v_2(x) + 1$, then after the next turn, the numbers of coins in the resulting piles are even and have different 2-adic valuations, and the smaller of the 2-adic valuations of the numbers of coins in the piles is equal to $v_2(x)$, which is even. This proves the claim. □

This establishes that Bob has a winning strategy if and only if a, b are both even and have the same 2-adic valuations, and this common 2-adic valuation is even. ■

Exercise 12 (Mexico National Olympiad 2020 P3, , proposed by Victor and Isaías de la Fuente)**.** Let $n \geq 3$ be an integer. Two players, Ana and Beto, play the following game. Ana tags the vertices of a regular n-gon with the numbers from 1 to n, in any order she wants. Every vertex must be tagged with a different number. Then, we place a turkey in each of the n vertices. These turkeys are trained for the following. If Beto whistles, each turkey moves to the adjacent vertex with greater tag. If Beto claps, each turkey moves to the adjacent vertex with lower tag. Beto wins if, after some number of whistles and claps, he gets to move all the turkeys to the same vertex. Ana wins if she can tag the vertices so that Beto can't do this. For each $n \geq 3$, determine which player has a winning strategy.

Walkthrough —

(a)

Solution 12. If n is even, then Ana wins for any arrangement of the tags. To see this, note that the vertices of the n-gon can be colored with two colors, say red and blue, such that adjacent vertices have different colors. Since n is even, there are the same number of red and blue vertices. After a whistle or a clap, the turkeys on the red vertices will move to blue vertices, and the turkeys on the blue vertices will move to red vertices.

■

Exercise 13 (Mexico National Olympiad 2020 P4, , proposed by Victor de la Fuente)**.** Let $n \geq 3$ be an integer. In a game there are n boxes in a circular array. At the beginning, each box contains an object which can be rock, paper or scissors, in such a way that there are no two adjacent boxes with the same object, and each object appears in at least one box. Same as in the game, rock crushes scissors, scissors cut paper, and paper covers rock. The game consists of moving objects from one box to another according to the following rule:

> Two adjacent boxes and one object from each one are chosen in such a way that these are different, and we move the loser object to the box containing the winner object. For example, if we picked rock from box A and scissors from box B, we move scissors to box A.

Prove that, applying the rule enough times, it is possible to move all the objects to the same box.

Walkthrough —

(a)

Solution 13. We claim that there are three consecutive boxes such that one of them contains a rock, one of the remaining two boxes contains a scissor, and the remaining box contains a paper. Consider a box $\mathcal{B}$ containing a rock. Assume that the above claim is false. Then the box next to $\mathcal{B}$ in the clockwise direction must contain a scissor or a paper. If it contains a scissor (resp. paper), then the box next to it contains only rocks, the next box contains only scissors (resp. paper), the next box contains only rocks, and moving this way in the clockwise direction, we will arrive at a box which contains a paper (resp. a scissor), proving the claim.

Consider three boxes, containing a rock, a scissor, and a paper in some order. We mark them, and called them a marked rock, or a marked scissor, or a marked paper. Note that each of a rock, a scissor, and a paper, wins against the stronger of the remaining two, and loses to the weaker of them. So, whichever of the first and the third box has a stronger marked element, the box in the middle capture it using its marked element. Next, the initially marked element of the middle square, moves to one of the first and the third box, which initially had a weaker marked element. Now, the only marked element from the middle square moves to the box having two marked elements. Hence, after a few moves, one can obtain a box, containing a rock, a scissor, and a paper. All the objects from the box to its left, that is, from the previous box in the anticlockwise direction, can be moved to this box. This proves that there is a box, containing at least a rock, a scissor, and a paper, so that one box to its left is empty.

Let $1 \leq k \leq n-2$ be a positive integer such that there is a box $\mathcal{C}$, containing at least a rock, a scissor, and a paper, and the k boxes to its left are empty. Let $\mathcal{D}$ denote the box to the right of the box $\mathcal{C}$, and O denote an object of $\mathcal{D}$. Note that $\mathcal{C}$ has two elements P, Q, such that O wins against the stronger of them, and loses to the weaker. Hence, the stronger of P, Q can be moved to $\mathcal{D}$, and the weaker one can be moved to $\mathcal{D}$ after that. Observe that $\mathcal{D}$ contains a rock, a scissor and a paper. Therefore, after a few moves, all objects of $\mathcal{C}$ can be moved to $\mathcal{D}$. Note that the $k+1$ boxes to the left of $\mathcal{D}$ are empty. By the principle of induction, it follows that all objects can be moved to a single box. ■

Exercise 14 (IMOSL 2020 C8, , Austria)**.** Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x+y$ and $|x-y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

1. one of the numbers on the blackboard is larger than the sum of all other numbers,
2. there are only zeros on the blackboard.

Player B must then give as many cookies to player A as there are numbers on the blackboard. Player A wants to get as many cookies as possible, whereas player B wants to give as few as possible. Determine the number of cookies that A receives if both players play optimally.

Exercise 15 (Brazil TST 2021 Test 2 Day 1 P1,)**.** Players A and B play a game on a blackboard that initially contains 2020 copies of the number 1. In every round, player A erases two numbers x and y from the blackboard, and then player B writes one of the numbers $x+y$ and $|x-y|$ on the blackboard. The game terminates as soon as, at the end of some round, one of the following holds:

1. one of the numbers on the blackboard is larger than the sum of all other numbers,
2. there are only zeros on the blackboard.

Player B has to pay to player A an amount in reais equivalent to the quantity of numbers left on the blackboard after the game ends. Show that player A can earn at least 8 reais regardless of the moves taken by B.

Gas Station Problem ISL 2021 C5

Exercise 16 (Canadian Junior Mathematical Olympiad 2022 P2,)**.** You have an infinite stack of T-shaped tetrominoes (composed of four squares of

side length 1), and an $n \times n$ board. You are allowed to place some tetrominoes on the board, possibly rotated, as long as no two tetrominoes overlap and no tetrominoes extend off the board. For which values of n can you cover the entire board?

> **Walkthrough** —
>
> **(a)**

Solution 14. Note that a 4×4 board can be covered by T-shaped tetrominoes, as shown in Fig. 9. Hence, an $n \times n$ board can be covered by T-shaped tetrominoes if n is a multiple of 4.

The following claim proves the converse.

> **Claim** — An $n \times n$ board can be covered by T-shaped tetrominoes only if n is a multiple of 4.

Proof of the claim. Assume that an $n \times n$ board can be covered by T-shaped tetrominoes. Each T-shaped tetromino covers exactly 4 unit squares, so the total number of unit squares covered must be a multiple of 4. Since the board has n^2 unit squares, it follows that n^2 must be a multiple of 4. This implies that n itself must be a multiple of 2.

Next, we will show that n is a multiple of 4. Consider the coloring of the board in a checkerboard pattern, where the top-left square is colored black. Each T-shaped tetromino covers either three black squares and one white square, or three white squares and one black square, depending on its orientation. Therefore, each T-shaped tetromino covers an odd number of squares of each color. Since n is even, the board has an even number of black squares and an even number of white squares. To cover the entire board, we would need to cover an even number of black squares and an even number of white squares, and hence, we would need to use an even number of T-shaped tetrominoes. This implies that the total number of squares covered, which is 4 times the number of tetrominoes, must be a multiple of 8. Therefore, n^2 must be a multiple of 8, which implies that n must be a multiple of 4. □

This proves that an $n \times n$ board can be covered by T-shaped tetrominoes if and only if n is a multiple of 4. ■

Exercise 17 (Junior Balkan MO 2023 P3, , proposed by Théo Lenoir, France)**.** Alice and Bob play the following game on a 100×100 grid, taking turns, with Alice starting first. Initially the grid is empty. At their turn, they choose an integer from 1 to 100^2 that is not written yet in any of the cells and choose an empty cell, and place it in the chosen cell. When there is no empty cell left, Alice computes the sum of the numbers in each row, and her score is the maximum of these 100 numbers. Bob computes the sum of the numbers in

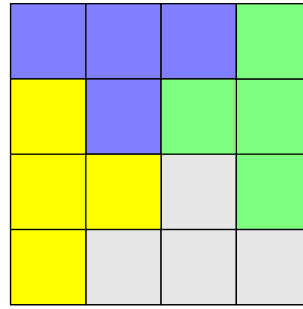


Figure 9: Canada Junior Mathematical Olympiad 2022 P2, Exercise 16

each column, and his score is the maximum of these 100 numbers. Alice wins if her score is greater than Bob's score, Bob wins if his score is greater than Alice's score, otherwise no one wins.

Find if one of the players has a winning strategy, and if so which player has a winning strategy.

Walkthrough —

(a)

Solution 15. Let us cover each of the 100 rows using 50 non-overlapping 1×2 dominoes. The two dominoes covering the four leftmost cells of the first row are called the *first* and *second* domino. We will show that Bob has a winning strategy, by using the following pairing strategy:

- If Alice places a number x in any cell of a domino other than the first and second domino, then Bob places $100^2 + 1 - x$ in the other cell of the same domino.
- If Alice places a number x in some cell of any of the first and second domino, then Bob places $100^2 + 1 - x$ in some cell of the other domino.

Note that Bob can always follow this strategy. Indeed, if Alice places a number x in a cell of a domino, then no cell of the grid filled earlier contains $100^2 + 1 - x$. Now, at the end of the game, the sum of the numbers in each row is $50 \cdot (100^2 + 1)$. On the other hand, the sum of the numbers in some column is at least $50 \cdot (100^2 + 1)$. Therefore, Bob's score is at least as large as Alice's score.

We will show that Bob's score is strictly greater than Alice's score. Let us assume for the sake of contradiction that Bob's score is equal to Alice's score, which is $50 \cdot (100^2 + 1)$. Then, the sum of the numbers in each column is $50 \cdot (100^2 + 1)$. In particular, the sum of the numbers in each of the first two

columns is $50 \cdot (100^2 + 1)$. Hence, the sum of the numbers in the first domino is

$$2 \cdot 50 \cdot (100^2 + 1) - 99 \cdot (100^2 + 1) = 100^2 + 1.$$

Note that Bob's strategy implies that the sum of the numbers in any of the first and second domino is not $100^2 + 1$. This is a contradiction, which completes the proof of the fact that Bob has a winning strategy. ■

> **Remark.** Dragomir Grozev argues that for the analogous problem on an $n \times n$ grid with n odd, Alice has a winning strategy.

Exercise 18 (Swiss Mathematical Olympiad 2023 Final Round Day 1 P2,). The wizard Albus and Brian are playing a game on a square of side length $2n + 1$ meters surrounded by lava. In the centre of the square there sits a toad. In a turn, a wizard chooses a direction parallel to a side of the square and enchants the toad. This will cause the toad to jump d meters in the chosen direction, where d is initially equal to 1 and increases by 1 after each jump. The wizard who sends the toad into the lava loses. Albus begins and they take turns. Depending on n, determine which wizard has a winning strategy.

> **Walkthrough** —
>
> **(a)**

Solution 16. Note that if $n = 0$, then Brian wins. Henceforth, we assume that $n \geq 1$. We claim that Brian wins if he follows the strategy of sending the toad in the opposite direction of Albus's previous jump.

Assume for the sake of contradiction that Brian's strategy fails after the kth turn of Albus. Since $n \geq 1$, we have $k \geq 2$. Note that a move of Albus and the subsequent move of Brian together cause a net displacement of the toad by 1 meter. Hence, two steps prior to Brian's kth turn, that is, after Brian's $(k-1)$th turn, the toad is at the edge of the square, otherwise, after next two moves, the toad would still be in the square. Moreover, the toad reaches the edge of the square after at least $2n$ moves, since the toad starts at the centre of the square. Therefore, $k - 1 \geq n$, and hence, $k \geq n + 1$. On the other hand, during k-th move of Albus, the toad jumps $2k$ meters from the edge of the square, which is more than the side length of the square, and hence, Albus loses. This is a contradiction, and hence, Brian's strategy works. ■

Exercise 19 (Columbia National Olympiad 2024 P1, , proposed by Juan David Restrepo). Daniel and Juan David have three piles of stones consisting of 30 yellow stones, 20 blue stones and 10 red stones respectively. They then take turns to remove one stone from any pile, starting with Daniel. Daniel wins if at any moment there are two piles with the same positive number of stones; in any other case, Juan David wins. Determine which of the two players has a winning strategy and describe it.

Walkthrough —

(a) Show that if Daniel removes a yellow stone during his turns, then he will win.

(b) Assume that after k turns in total, only one yellow stone is left.

(c) Assume that among these k turns, Daniel has removed x yellow stones, and Juan David has removed y yellow stones, b blue stones and r red stones.

(d) Show that $b + r \leq b + r + y \leq x \leq x + y = 29$.

(e) Prove that the difference between the number of yellow stones and the number of blue stones or the difference between the number of yellow stones and the number of red stones is less than or equal to zero after k turns.

(f) Conclude that Daniel wins.

Solution 17. The first player, Daniel has a winning strategy. The strategy of Daniel is to remove a yellow stone during his turns. We claim that Daniel will win by following this strategy. More precisely, we will show that Daniel will win when only one yellow stone is left or earlier. Assume that after k turns in total, only one yellow stone is left. Assume that among these k turns, Daniel has removed x yellow stones, and Juan David has removed y yellow stones, b blue stones and r red stones. Note that after these k turns, there are $30 - x - y = 1$ yellow stones, $20 - b$ blue stones and $10 - r$ red stones. Since Daniel is the first player, it follows that at any stage of the game, the number of turns of Juan David is either equal to or one less than the number of turns of Daniel. This implies that $b + r \leq b + r + y \leq x \leq x + y = 29$. This shows that at least one of the inequalities $b \leq 19$, $r \leq 9$ holds. This gives that after k turns, at least one of the differences

$$\text{the number of yellow stones} - \text{the number of blue stones} = b - 19,$$
$$\text{the number of yellow stones} - \text{the number of red stones} = r - 9$$

is less than or equal to zero. Observe that these two differences were both positive at the beginning of the game. Therefore, at some moment during the game, within these k turns, one of these two differences must have become zero. In other words, at some moment during the game, one of the red and blue piles had the same positive number of stones as the yellow pile. Noting that during these k turns, the number of yellow stones is always positive, we conclude that Daniel wins. ■

Exercise 20 (Columbia Junior Mathematical Olympiad 2024 P3, , proposed by Santiago Rodriguez)**.** Let $n > 1$ be a positive integer. There are n rocks arranged in a circle, with one frog sitting on each rock. The numbers $1, 2, \ldots, n$ are written on the rocks in some order. After a signal is given, each

frog looks at the number i written on the rock it occupies and then jumps i positions clockwise along the circle. Determine all values of n for which it is possible to assign the numbers $1, 2, \ldots, n$ to the rocks in such a way that, after the signal, no two frogs land on the same rock.

Walkthrough —

(a) Label the rocks as the first, second, ..., n-th rock in the clockwise order. For each $i \in \{1, 2, \ldots, n\}$, let a_i be the number written on the i-th rock.

(b) Observe that if n is a positive integer such that there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal, then the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are distinct modulo n.

(c) Show that integer n satisfying the given condition must be odd.

(d) Show that if n is odd, then there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal.

Solution 18. Let $n > 1$ be a positive integer such that there is an assignment of the numbers $1, 2, \ldots, n$ to the rocks such that no two frogs land on the same rock after the signal. Label the rocks as the first, second, ..., n-th rock in the clockwise order. For each $i \in \{1, 2, \ldots, n\}$, let a_i be the number written on the i-th rock. Then, after the signal, the frog on the i-th rock jumps to the $(i + a_i)$-th rock (where the integer $i + a_i$ is taken modulo n). Since no two frogs land on the same rock, the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are distinct modulo n. In particular, the numbers $i + a_i$ for $i = 1, 2, \ldots, n$ are congruent to the numbers $1, 2, \ldots, n$ in some order modulo n. Therefore, we have

$$(1 + a_1) + (2 + a_2) + \cdots + (n + a_n) \equiv 1 + 2 + \cdots + n \pmod{n},$$

which implies that $a_1 + a_2 + \cdots + a_n \equiv 0 \pmod{n}$. Since

$$a_1 + a_2 + \cdots + a_n = 1 + 2 + \cdots + n = \frac{n(n+1)}{2},$$

it follows that n divides $\frac{n(n+1)}{2}$. This implies that $n + 1$ is even, so n is odd.

Conversely, let $n > 1$ be an odd positive integer. Consider the assignment of the numbers $1, 2, \ldots, n$ to the rocks in the order $1, 2, \ldots, n$ in the clockwise direction. Then, after the signal, the frog on the i-th rock jumps to the $(i+i)$-th rock, which is the $(2i)$-th rock modulo n. Since n is odd, the numbers $2i$ for $i = 1, 2, \ldots, n$ are distinct modulo n. Therefore, no two frogs land on the same rock after the signal.

Hence, the integers $n > 1$ for which it is possible to assign the numbers $1, 2, \ldots, n$ to the rocks in such a way that no two frogs land on the same rock after the signal are precisely the odd integers. ■

Exercise 21 (Taiwan TST 2024 Round 1 Quiz 1, , proposed by Ho-Chien Chen)**.** Let $n \geq 5$ be a positive integer. There are n stars with values 1 to n, respectively. Anya and Becky play a game. Before the game starts, Anya places the n stars in a row in whatever order she wishes. Then, starting from Becky, each player takes the left-most or right-most star in the row. After all the stars have been taken, the player with the highest total value of stars wins; if their total values are the same, then the game ends in a draw. Find all n such that Becky has a winning strategy.

Walkthrough —

(a)

Solution 19. We claim that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. If $n \equiv 2 \pmod 4$, then compares the sum of the values of the stars in the odd positions, and the sum of the values of the stars in the even positions. Since $n \equiv 2 \pmod 4$, one of these sums is strictly greater than the other. If the sum of the values of the stars in the odd positions is greater than the sum of the values of the stars in the even positions, then Becky always takes the stars in the odd positions. Otherwise, Becky always takes the stars in the even positions. In either case, Becky wins.

If $n = 2m+1$ with $m \geq 2$, then Anya places the stars with values $1, 2, \ldots, m+1$ in the odd positions from left to right, and places the remaining stars in the even positions from left to right. Then, no matter how Becky plays, Anya takes the adjacent star, which is in an even position. Note that the sum of the values of the stars in the odd positions is

$$1 + 2 + \cdots + (m+1) = \frac{(m+1)(m+2)}{2},$$

and the sum of the values of the stars in the even positions is

$$(m+2) + (m+3) + \cdots + (2m+1) = \frac{3m}{2}(m+1).$$

Since $m \geq 2$, we have

$$\frac{3m}{2}(m+1) > \frac{(m+1)(m+2)}{2},$$

so Anya wins.

Assume that $n = 4m + 4$ with $m \geq 1$. Then Anya uses a similar strategy to force a draw or a win for Anya. First, Anya places the stars with values $m+1, m+2, \ldots, 2m$ in the first m odd positions from left to right, the stars with values $2m+1, 2m+2, \ldots, 3m$ in the next m odd positions from left to right, the stars with values $3m+1, 3m+2, \ldots, 4m$ in the first m even positions

from left to right, and the stars with values $1, 2, \ldots, m$ in the next m even positions from left to right. In other words, Anya places the stars in the order

$$m+1, 3m+1, m+2, 3m+2, m+3, 3m+3, \ldots, 2m, 4m,$$
$$2m+1, 1, 2m+2, 2, \ldots, 3m, m$$

from left to right. Anya play by taking the star adjacent to the star taken by Becky in each turn.

Claim — As a consequence of the above strategy, the game ends in a draw or a win for Anya.

Proof of the Claim. Let us first consider the case when Becky takes more or an equal number of stars from the left end of the row than from the right end of the row. Note that Anya takes the stars with values $3m+1, 3m+2, \ldots, 4m$. As per Anya's strategy,

$$\begin{aligned}
&\text{the sum of the values of stars taken by Anya} \\
&\quad - \text{the sum of the values of stars taken by Becky} \\
&\geq (3m+1) + (3m+2) + \cdots + (4m) \\
&\quad - ((m+1) + (m+2) + \cdots + (2m)) \\
&\quad - |(2m+1) - 1| - |(2m+2) - 2| - \cdots - |3m - m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}$$

Next, we consider the case when Becky takes more stars from the right end of the row than from the left end of the row. Note that Anya takes the stars with values $2m+1, 2m+2, \ldots, 3m$. As per Anya's strategy,

$$\begin{aligned}
&\text{the sum of the values of stars taken by Anya} \\
&\quad - \text{the sum of the values of stars taken by Becky} \\
&\geq (2m+1) + (2m+2) + \cdots + (3m) \\
&\quad - (1 + 2 + \cdots + m) \\
&\quad - |(3m+1) - (m+1)| - |(3m+2) - (m+2)| - \cdots - |4m - 2m| \\
&= 2m^2 - 2m^2 \\
&= 0.
\end{aligned}$$

This proves the claim. □

This proves that Becky has a winning strategy if and only if $n \equiv 2 \pmod 4$. ■

The content posted here and at this blog by Evan Chen are quite useful.

Exercise 22 (Mexico National Olympiad 2024 P6,). Ana and Beto play on a blackboard where all integers from 1 to 2024 (inclusive) are written. On each turn, Ana chooses three numbers a, b, c written on the board and then Beto erases them and writes one of the following numbers:

$$a + b - c, a - b + c, -a + b + c.$$

The game ends when only two numbers are left on the board and Ana cannot play. If the sum of the final numbers is a multiple of 3, Beto wins. Otherwise, Ana wins. Who has a winning strategy?

Walkthrough —

(a) Can Beto have a strategy to keep the parity of the number of integers congruent to i modulo 3 unchanged for any $i \in \{0, 1, 2\}$?

(b) Note that if Ana chooses three numbers which are congruent to $0, 1, 2$ modulo 3 in some order, then Beto will fail to do so, and otherwise Beto keep these parities unchanged.

(c) If Ana makes such a move, then Beto can write the integer among the three choices that is divisible by 3.

(d) For the next moves, can Beto follow some strategy to keep the parity of the number of integers congruent to i modulo 3 unchanged for any $i \in \{0, 1, 2\}$, and keep at least two integers divisible by 3 to ensure winning the game?

(e) Note that if Ana chooses three numbers which are congruent to any of the unordered triples $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$ modulo 3 in some order, then Beto will again fail to do so, and otherwise Beto can have a strategy as sought in the previous step.

(f) If Ana makes such a move, then does Beto have a strategy to reduce the game to a collection of an even number of integers congruent to 0 modulo 3, an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3, similar to the initial configuration of the game, and then start afresh to win the game by induction?

Solution 20. Let us describe two rules. Next, we provide a strategy of Beto based on these two rules, and then we verify that the strategy is winning for Beto.

The first rule is that

(i) if the three numbers chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then Beto writes the number that is congruent to 0 modulo 3,

(ii) if the three numbers chosen by Ana are not pairwise distinct modulo 3, then Beto writes a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

The second rule is that

(i) if the three numbers chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then Beto writes the number that is congruent to 0 modulo 3,

(ii) if the three numbers chosen by Ana are congruent to $0, 0, 1$ (resp. $0, 0, 2$) modulo 3 in some order, then Beto writes the number that is congruent to 2 (resp. 1) modulo 3,

(iii) if the three numbers chosen by Ana are not congruent to any of the above-mentioned unordered triples (that is, to any of $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$), then Beto writes a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

The strategy of Beto is to follow the first rule in the first move. For any subsequent move, Beto follows the same rule as in the previous move if the parity of the number of integers congruent to i modulo 3 did not change in the previous move for any $i \in \{0, 1, 2\}$, and otherwise Beto follows the other rule.

Claim — The first rule is well-defined.

Proof of the claim. Note that if three integers x, y, z are congruent to $0, 1, 2$ modulo 3, then $x + y + z$ is congruent to 0 modulo 3, and using the congruences

$$\begin{aligned} x + y - z &\equiv -2z \pmod 3, \\ x - y + z &\equiv -2y \pmod 3, \\ -x + y + z &\equiv -2x \pmod 3, \end{aligned}$$

it follows that exactly one of $x + y - z, x - y + z, -x + y + z$ is congruent to 0 modulo 3.

Also note that if three integers x, y, z are not pairwise distinct modulo 3, then some two of x, y, z are congruent to each other modulo 3, and the other integer among x, y, z is congruent to at least one of $x + y - z, x - y + z, -x + y + z$ modulo 3. Therefore, Beto can write a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$. Hence, the first rule is well-defined. □

Claim — The second rule is well-defined.

Proof of the claim. By the first part of the previous proof, it follows that if the three integers a, b, c chosen by Ana are congruent to $0, 1, 2$ modulo 3 in some order, then there is a unique number among $a + b - c, a - b + c, -a + b + c$ that is congruent to 0 modulo 3, and hence Beto can write that number.

The content posted here and at this blog by Evan Chen are quite useful.

If the three integers a, b, c chosen by Ana are congruent to $0, 0, 1$ modulo 3 in some order, then the integers $a+b-c, a-b+c, -a+b+c$ are congruent to

$$\begin{aligned}(0+0+1)-2\times 1 &\equiv 2 \pmod 3,\\ (0+0+1)-2\times 0 &\equiv 1 \pmod 3,\\ (0+0+1)-2\times 0 &\equiv 1 \pmod 3,\end{aligned}$$

in some order, and hence Beto can write the number that is congruent to 2 modulo 3.

Similarly, if the three integers a, b, c chosen by Ana are congruent to $0, 0, 2$ modulo 3 in some order, then the integers $a+b-c, a-b+c, -a+b+c$ are congruent to

$$\begin{aligned}(0+0+2)-2\times 2 &\equiv 1 \pmod 3,\\ (0+0+2)-2\times 0 &\equiv 2 \pmod 3,\\ (0+0+2)-2\times 0 &\equiv 2 \pmod 3,\end{aligned}$$

in some order, and hence Beto can write the number that is congruent to 1 modulo 3.

Finally, if the three integers a, b, c chosen by Ana are not congruent to any of the above-mentioned unordered triples (that is, to any of $(0, 1, 2)$, $(0, 0, 1)$, $(0, 0, 2)$), then the three integers a, b, c are not pairwise distinct modulo 3, and some two of a, b, c are congruent to each other modulo 3, and the other integer among a, b, c is congruent to at least one of $a+b-c, a-b+c, -a+b+c$ modulo 3. Therefore, Beto can write a number so that the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$.

This proves that the second rule is well-defined. □

Claim — The strategy of Beto is winning for Beto if the game is played on any collection of at least two integers, where there are an even number of integers congruent to 0 modulo 3, an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3.

Proof of the claim. Note that if the game is played on a collection of two integers, then the integers on the board are congruent to 1 and 2 modulo 3 in some order, or both integers on the board are congruent to 0 modulo 3, and hence, their sum is congruent to 0 modulo 3, and Beto wins. Let $k \geq 2$ be an integer, and assume that the above strategy of Beto is winning for Beto if the game is played on any collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3.

Suppose that the game is played on a collection of $k+1$ integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3.

If Beto follows part (ii) of the first rule in the first move, then the parity of the number of integers congruent to i modulo 3 does not change for any $i \in \{0, 1, 2\}$ in the first move, and this reduces the game to a collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3, and hence Beto wins by the induction hypothesis. Therefore, we may assume that Beto follows part (i) of the first rule in the first move.

Note that after the first move, there are an even number of integers congruent to i modulo 3 for any $i \in \{0, 1, 2\}$, and at least two integers which are congruent to 0 modulo 3. If the game ends here, then the sum of the remaining integers is congruent to 0 modulo 3, and Beto wins. It remains to consider the case that the game does not end here.

If until the end of the game, Beto follows part (iii) of the second rule, then the parity of the number of integers congruent to i modulo 3 remain even for any $i \in \{0, 1, 2\}$ throughout the game, and noting that in after an application of part (iii) of the second rule, the number of integers congruent to 0 modulo 3 is at least two, it follows that the sum of the integers left on the board at the end of the game is congruent to 0 modulo 3, and Beto wins. If prior to the end of the game, Beto follows part (i) or part (ii) of the second rule at least once, then after the first application of part (i) or part (ii) of the second rule, there are an even number of integers congruent to 0 modulo 3, and an odd number of integers congruent to 1 modulo 3, and an odd number of integers congruent to 2 modulo 3. This reduces the game to a collection of at least two and at most k integers, satisfying the above-mentioned conditions on the number of integers congruent to $0, 1, 2$ modulo 3, and hence Beto wins by the induction hypothesis. □

■

Exercise 23 (Dutch IMO TSTST 2024 P4,). Let n be a positive integer with $n \geq 3$. Consider a board of $n \times n$ boxes. In each step taken, the colors of the 5 boxes that make up the figure below change color (black boxes change to white and white boxes change to black). The figure can be rotated $90°, 180°$ or $270°$. Initially, all the boxes are white. Determine for what values of n it can be achieved, through a series of steps, that all the squares on the board are black.

> **Walkthrough** —
>
> **(a)**

Solution 21. to write ■

Exercise 24 (Austrian Mathematical Olympiad, National Competition, Preliminary Round 2025 P3, , proposed by Michael Reitmeir). Consider the

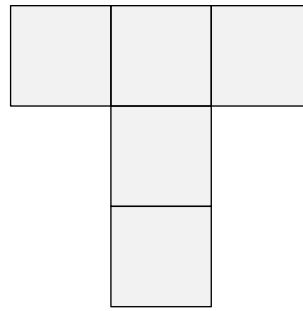


Figure 10: Dutch IMO TSTST 2024 P4, Exercise 23

following game for a positive integer n. Initially, the numbers $1, 2, \ldots, n$ are written on a board. In each move, two numbers are selected such that their difference is also present on the board. This difference is then erased from the board. (For example, if the numbers $3, 6, 11$ and 17 are on the board, then 3 can be erased as $6 - 3 = 3$, or 6 as $17 - 11 = 6$, or 11 as $17 - 6 = 11$.)

For which values of n, is it possible to end with only one number remaining on the board?

Walkthrough — Assume that it is possible to end with only one number remaining on the board.

(a) Show that if d is an odd integer dividing the integers present on the board at some stage, then d divides the integer that was erased in the move prior to this stage, that is, d divides all the integers present on the board prior to this stage as well.

(b) Show that n has no odd positive divisor other than 1.

Next, assume that n is a power of 2. Use induction to show that it is possible to end with only one number remaining on the board.

Solution 22. By the claims below, it follows that it is possible to end with only one number remaining on the board if and only if n is a power of 2.

Claim — Let n be a positive integer. Assume that it is possible to end with only one number remaining on the board. Then n is a power of 2.

Proof of the Claim. Let d be an odd positive divisor of n. We will show that d must be 1. Since no two of the numbers $1, 2, \ldots, n$ differ by n, the number n cannot be erased in any move. Therefore, n must be the last remaining number on the board.

Assume that d divides all the integers present on the board at some stage in the game. Denote these integers by $a_1, a_2, \ldots, a_k$. Denote by b the integer that was erased in the move prior to this stage.

Consider the case that b is the difference of b and a_i for some i. Then b is equal to one of $b - a_i, a_i - b$. Since a_i is positive, we obtain $b = a_i - b$, which implies $a_i = 2b$. Since d divides a_i and d is odd, it follows that d divides b.

Now, consider the case that b is equal to $a_i - a_j$ for some i and j with $i \neq j$. Since d divides both a_i and a_j, it follows that d divides b.

This proves that if d divides all the integers present on the board at some stage, then d divides the integer that was erased in the move prior to this stage, that is, d divides all the integers present on the board prior to this stage as well. Since d divides n, it follows that d divides all the integers present on the board at every stage prior to the last stage. In particular, d divides 1, which implies $d = 1$.

This proves that n has no odd positive divisor other than 1. Therefore, n is a power of 2. □

Claim — Let n be a positive integer. Assume that n is a power of 2. Then it is possible to end with only one number remaining on the board.

Proof of the Claim. Write $n = 2^k$ with $k \geq 0$. Using induction, we will prove the claim. For the base case $k = 0$, we have $n = 1$. In this case, we already have only one number remaining on the board, so the claim holds. Also note that the claim holds for $k = 1$ as well, since we can erase 1 and be left with 2.

Assume that the claim holds for some $k \geq 1$, that is, for $n = 2^k$. Consider $n = 2^{k+1}$. We can divide the numbers $1, 2, \ldots, 2^{k+1}$ into two groups: $1, 2, \ldots, 2^k$ and $2^k + 1, 2^k + 2, \ldots, 2^{k+1}$. Note that

$$2^k + i = 2^{k+1} - (2^k - i)$$

for any i with $1 \leq i \leq 2^k - 1$. Therefore, we can erase the numbers $2^k + 1, 2^k + 2, \ldots, 2^{k+1} - 1$ by using the numbers $1, 2, \ldots, 2^k - 1$ and 2^{k+1}. Thus, we can reduce the numbers on the board to $1, 2, \ldots, 2^k$ and 2^{k+1}. By the induction hypothesis, we can reduce $1, 2, \ldots, 2^k$ to 2^k. Finally, when we are left with 2^k and 2^{k+1} on the board, we can erase 2^k and be left with 2^{k+1} as the only number remaining on the board. By induction, the claim holds for all $k \geq 0$. □

■

Exercise 25 (Rioplatense Mathematical Olympiad 2025 Level 1 Grade 8-9 P3,). Liko plays a game called The 24-N cards. In this game, N cards are placed on the table, each showing an integer between 1 and 9, inclusive. Some cards may have the same number. Liko wins if she can write a mathematical expression equal to 24, using each of the cards exactly once. She may only use the operations addition, subtraction, multiplication, and division, and she may group the operations in any way she wants, but multiplying by 0 is not allowed

in this game. For instance, if the cards on the table are $\{7, 1, 7, 2\}$, then Liko can win by forming

$$(7 \times 7 - 1) \div 2 = 24.$$

Determine whether there exists a number N such that, regardless of what numbers appear on the N cards, Liko can always win.

> **Walkthrough** —
>
> **(a)**

Solution 23. We claim that for $N = 2 \times 68 + 9$, Liko can always win, regardless of the numbers on the cards. Given $2 \times 68 + 9$ numbers between 1 and 9, we can group them into at least 68 pairs of numbers, such that each pair consists of two equal numbers, and the remaining at most 9 numbers are pairwise distinct. These pairs of equal numbers can be used to create 68 copies of the number 1 by dividing the two equal numbers in each pair. Then, we have at least 68 copies of the number 1 and at most 8 pairwise distinct numbers between 2 and 9. Note that the sum of these at most 8 distinct numbers is at most $2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 = 44$. Using at most 44 of the 68 copies of 1, we can reduce the sum of those at most 8 distinct numbers to zero. Thus, we are left with at least $68 - 44 = 24$ copies of the number 1. Finally, we can add 24 of these 1's together to get 24. Next, we multiply 24 by the remaining numbers (which are all equal to 1), if any, to get 24 again. ■

Exercise 26 (Brazil National Olympiad 2025 Level 2 P2,). Let m be a positive integer. Ana and Banana play the following game on a 5×5 board, which is initially filled with 0's in all squares. Taking turns starting with Ana, they choose a square on the board and add an integer from the set $\{1, 2, 3, 4, 5\}$ to the number currently in that square, such that the number in the square does not exceed m. The winner is the player who makes a move that completes a row, a column, or one of the two main diagonals with the number m in all five squares. For which positive integers m does Ana, the first player, have a winning strategy?

> **Walkthrough** —
>
> **(a)**

Solution 24.

> **Claim** — If m is a multiple of 6, then Banana has a winning strategy.

Proof of the claim. □

Claim — If m is not a multiple of 6, then Ana has a winning strategy.

Proof of the claim. □

■

Exercise 27 (Columbia National Olympiad 2025 P2, , proposed by Santiago Rodriguez)**.** The numbers $1, 2, \ldots, 2025$ are written around a circle in some order. On each turn, Celeste chooses an integer $1 \leq n \leq 2025$. Then she selects the number n and the next $n-1$ numbers following it in the clockwise direction and inverts their order on the circle. Prove that, after a finite amount of turns, Celeste can put the numbers on the circle in the order $1, 2, 3, \ldots, 2025$ in the clockwise direction, regardless of their original arrangement on the circle.

Walkthrough —

(a) Show that if the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the order of the other numbers unchanged.

(b) Show that if the numbers a, b appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a in the clockwise direction, while keeping the order of the other numbers unchanged.

(c) Using the above two results, prove that, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, after a finite number of turns, they can be arranged in the order $1, 2, 3, \ldots, 2025$ in this order in the clockwise direction.

Solution 25. Let us prove a few claims about the operation.

Claim — If the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 2$ to the three numbers $2, 3, x$ to get the arrangement $3, 2, x$. Then we apply the operation with $n = 3$ to the three numbers $3, 2, x$ to get the arrangement $x, 2, 3$. Note that the order of the other numbers is unchanged in both operations. □

Claim — Let a, b, c be three distinct integers among $1, 2, \ldots, 2025$ different from $2, 3$. If the numbers $2, 3, a, b, c$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be

arranged in the order $2, 3, b, a, c$ in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 3$ to the arrangement $2, 3, a, b, c$ to get the arrangement $2, b, a, 3, c$. Then we apply the operation with $n = 2$ to the arrangement $2, b, a, 3, c$ three times to get the arrangement $b, a, 3, 2, c$. Next, we apply the operation with $n = 3$ to the arrangement $b, a, 3, 2, c$ to get the arrangement $b, a, c, 2, 3$. Note that the order of the other numbers is unchanged in all operations. Applying the prior claim 2020 times, we get the arrangement $2, 3, b, a, c$, without changing the order of the other numbers. □

Claim — Let a, b, c be three distinct integers among $1, 2, \ldots, 2025$ different from $2, 3$. If the numbers a, b, c appear consecutively in this order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a, c in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. Applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction, we obtain $2, 3$ in this order in the clockwise direction. Then we can apply the previous claim to $2, 3, a, b, c$ to get the arrangement $2, 3, b, a, c$. Next, we apply the first claim to $2, 3$ and the integer next to 3 in the clockwise direction to place $2, 3$ in this order in the clockwise direction so that 3 is placed in the same position as before. Finally, we move 2 back to its original position by applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction. Note that the order of the other numbers is unchanged in all operations. □

Now, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, we can arrange $2, 3$ in this order in the clockwise direction. Next, applying the previous claim repeatedly, we obtain the arrangement $1, 2, 3$ in this order in the clockwise direction. Finally, we can place the remaining numbers $4, 5, \ldots, 2025$ in the correct order by repeatedly applying the previous claim to adjacent triplets of numbers that are out of order. ■

Here is an alternative solution to the problem.

Solution 26. Let us prove the following claims.

Claim — If the numbers $2, 3, x$ appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order $x, 2, 3$ in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. We can apply the operation with $n = 2$ to the three numbers $2, 3, x$ to get the arrangement $3, 2, x$. Then we apply the operation with $n = 3$ to the three numbers $3, 2, x$ to get the arrangement $x, 2, 3$. Note that the order of the other numbers is unchanged in both operations. □

> **Claim** — If the numbers a, b appear consecutively in that order in the clockwise direction, then after a finite number of turns, they can be arranged in the order b, a in the clockwise direction, while keeping the order of the other numbers unchanged.

Proof of the Claim. Applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction, we can place $2, 3$ in this order in the clockwise direction. Then applying the prior claim to $2, 3$ and the integer next to 3 in the clockwise direction, we can obtain the arrangement $2, 3, a$ in this order in the clockwise direction. Next, applying the operation with $n = 3$ to the arrangement $2, 3, a, b$ gives the arrangement $2, b, a, 3$. Then we apply the operation with $n = 2$ to the arrangement $2, b, a, 3$ two times to get the arrangement $b, a, 2, 3$. Applying the prior claim to $2, 3$ and the integer next to 3 in the clockwise direction, we can place $2, 3$ in this order in the clockwise direction so that 3 is placed in the same position as before. Finally, we move 2 back to its original position by applying the operation with $n = 2$ to 2 and the integer next to 2 in the clockwise direction. Note that the order of the other numbers is unchanged in all operations. □

Now, given any arrangement of the numbers $1, 2, \ldots, 2025$ around the circle, we can arrange $2, 3$ in this order in the clockwise direction. Next, applying the previous claim a finite number of times, to integers lying between 3 and 4 in the clockwise direction, we can obtain the arrangement $2, 3, 4$ in this order in the clockwise direction. Next, after obtaining the arrangement $2, 3, 4, \ldots, k$ in this order in the clockwise direction, for some $4 \leq k < 2025$, we can apply the previous claim a finite number of times to integers lying between k and $k + 1$ in the clockwise direction to get the arrangement $2, 3, 4, \ldots, k, k + 1$ in this order in the clockwise direction. It follows that we can obtain the arrangement $2, 3, 4, \ldots, 2025$ in this order in the clockwise direction. This gives the desired arrangement $1, 2, 3, \ldots, 2025$ in this order in the clockwise direction. ■

Exercise 28 (All-Russian Mathematical Olympiad 2025 Grade 9 Day 2 P7, , proposed by I. A. Efremov)**.** The numbers $1, 2, 3, \ldots, 60$ are written in a row in that exact order. Igor and Ruslan take turns inserting the signs $+, -, \times$ between them, starting with Igor. Each turn consists of placing one sign. Once all signs are placed, the value of the resulting expression is computed. If the value is divisible by 3, Igor wins; otherwise, Ruslan wins. Which player has a winning strategy regardless of the opponent's moves?

Walkthrough —

(a) Igor, the first player, has a winning strategy.

(b) Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively.

(c) Igor in his first turn places a $-$ sign between 30 and 31.

(d) After that, the numbers $1, 2, 3, \ldots, 30$ can be thought as grouped together and the numbers $31, 32, 33, \ldots, 60$ can also be thought as grouped together.

(e) For each choice of Ruslan, Igor can respond by placing the corresponding sign in the other half.

Solution 27. We claim that Igor, the first player, has a winning strategy. Note that the numbers $1, 2, 3, \ldots, 30$ are congruent to $31, 32, 33, \ldots, 60$ modulo 3, respectively. Igor in his first turn places a $-$ sign between 30 and 31. Next, after each turn of Ruslan, which is to place a sign in one of the two halves, that is, to place a sign between two consecutive numbers between 1 and 30 or to place a sign between two consecutive numbers between 31 and 60, Igor can respond by placing the corresponding sign in the other half. Here the corresponding sign means that if Ruslan places a $\times$ sign, then Igor also places a $\times$ sign, and if Ruslan places a $+$ sign, then Igor places a $-$ sign, and if Ruslan places a $-$ sign, then Igor places a $+$ sign. After all signs are placed, the resulting expression can be written as the difference of two expressions, one of which consists of the numbers $1, 2, 3, \ldots, 30$ and the other of which consists of the numbers $31, 32, 33, \ldots, 60$. The strategy of Igor ensures that these two expressions are congruent modulo 3. Therefore, Igor wins. ■

Exercise 29 (Cono Sur Olympiad 2025 P4,). Lucero and Pablo play a game on the board shown in Fig. 11. Lucero plays first, and they take turns. In each turn, a player chooses an unpainted circle from the bottom row and paints it blue, green, or red. This continues for four turns until the entire bottom row is painted.

Then, the rest of the board is painted according to the following rules:

(i) If two adjacent circles in a row are the same color, the circle above and adjacent to them is painted that same color.

(ii) If two adjacent circles in a row are of different colors, the circle above and adjacent to them is painted with the third color.

This procedure is repeated until all the circles on the board are painted. Lucero wins if the single top circle is painted red or green, and Pablo wins if it is painted blue.

Determine who has a winning strategy.

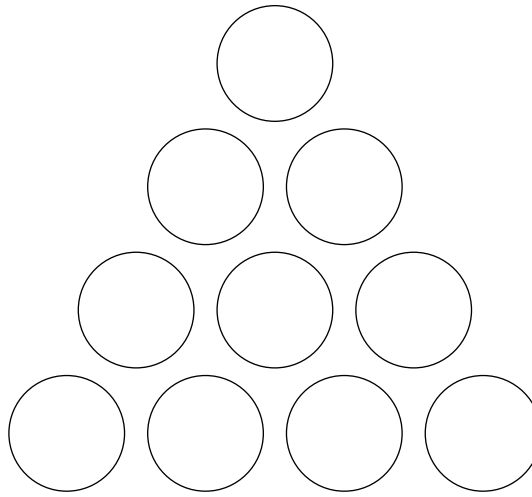


Figure 11: Cono Sur Olympiad 2025 P4, Exercise 29

Walkthrough —

(a) Use the congruence classes modulo 3 to represent the colors of the circles. For example, we can represent blue as 0, green as 1, and red as -1 modulo 3.

(b) Let a, b, c, d be the colors of the circles in the bottom row from left to right, respectively, represented as congruence classes modulo 3.

(c) Express the color of the top circle as a function of the colors of the circles in the bottom row.

(d) Determine a winning strategy for Pablo based on the colors of the circles in the bottom row.

Solution 28. We will show that Pablo has a winning strategy.

Let us consider the colors as congruence classes modulo 3 as follows: 0 for blue, 1 for green, and -1 for red. We can represent the colors of the circles in the bottom row, from left to right, as a sequence of four congruence classes, which we can denote as a, b, c, d, where each of a, b, c, d is in $\{0, 1, -1\}$. Note that if the colors of two adjacent circles are x, y modulo 3, then the color of the circle above them is $-(x + y)$ modulo 3. Indeed, if the colors of two adjacent circles are x, y modulo 3, and x, y are congruent modulo 3, then the color of the circle above them is x modulo 3, which is $-(x + y)$ modulo 3 since $x \equiv y \pmod 3$. If x, y are not congruent modulo 3, then the color of the circle above them is the third color, which is $-(x+y)$ modulo 3. Therefore, the color of the top circle can be expressed as a function of a, b, c, d as follows. In fact, note the colors of the circle in the third row are $-(a + b)$, $-(b + c)$, and $-(c + d)$

modulo 3. Then, the colors of the circles in the second row are

$$-(-(a+b)+-(b+c)) = a+2b+c,$$
$$-(-(b+c)+-(c+d)) = b+2c+d$$

modulo 3. Finally, the color of the top circle is

$$-(a+2b+c+b+2c+d) = -a-3b-3c-d \equiv -a-d \pmod 3.$$

Therefore, the color of the top circle is determined by the colors of the leftmost and rightmost circles in the bottom row.

If Lucero, in her first turn, chooses to color the leftmost or the rightmost circle in the bottom row, then Pablo colors the circle on the opposite end of the bottom row with its negative congruence class modulo 3. This ensures that the color of the top circle is 0 modulo 3, which means that the top circle is blue, and Pablo wins.

If Lucero, in her first turn, chooses to color any of the two middle circles in the bottom row, then Pablo colors the other middle circle with any color. In next turn, Lucero colors one of the two remaining circles in the bottom row, and Pablo colors the other remaining circle in the bottom row with its negative congruence class modulo 3. This again ensures that the color of the top circle is 0 modulo 3, which means that the top circle is blue, and Pablo wins.

Therefore, Pablo has a winning strategy, and the answer is Pablo. ■

Exercise 30 (Singapore Junior Mathematical Olympiad 2025 P3,). Jack and Jill play the following game: Jack throws 3 dice and Jill can select some of them, possibly none, and turn each of them to the opposite side. Jill wins if the sum of the values on the dice is a multiple of 4. Can Jill always win? (Note the game is played with standard dice where the sum of the numbers on opposite sides is 7.)

> **Walkthrough** —
>
> **(a)**

Solution 29. Note that if the number on the top face of a die is is 1 or 2 modulo 4, that is, if it is 1, 2, 5 or 6, then for any $i \in \{1,2\}$, the die can be flipped (if required) to make the value on the top face congruent to i modulo 4. Also if the number on the top face of a die is 3 or 0 modulo 4, that is, if it is 3 or 4, then for any $i \in \{0,3\}$, the die can be flipped (if required) to make the value on the top face congruent to i modulo 4. Jill can always win by the following strategy:

- If the values on the top faces of all the dice lie in $\{1,2,5,6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $1,1,2$ modulo 4.

- If the values on the top faces of exactly two dice lie in $\{1, 2, 5, 6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $2, 2, 4$ modulo 4.
- If the values on the top faces of exactly one die lie in $\{1, 2, 5, 6\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $1, 3, 4$ modulo 4.
- If the values on the top faces of all the dice lie in $\{3, 4\}$, then Jill can flip some of the dice to make the values on the top faces congruent to $4, 4, 4$ modulo 4.

Using the above strategy, Jill can make the sum of the values on the top faces of the dice congruent to 0 modulo 4, and hence win the game. ■

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)